

There is hence a tradeoff between the higher residuals and the higher transaction costs for trades on the consecutive forward rate curve. Altogether, using consecutive forward rates is particularly advisable in case the statistical properties of the PCA need to be improved and in case the profit from asset selection (residuals) is high.

In general terms, once an investor has defined the desired factor exposure of his portfolio, the PCA residuals to these factors will show the best way to get this exposure. And since the residuals are uncorrelated to the factors, this asset selection method offers a source of profit which is uncorrelated to the factors (i.e. to whether the view on factors turns out to be right or wrong).

For example, imagine an investor is bullish on Bunds in general and thus decides to buy a Bund¹⁷ (i.e. to get exposure to factor 1). He can then look at the chart shown in Figure 3.17 of one-factor residuals (i.e. the difference between actual Bund yields and where they typically should be, given the overall yield level). (Note again how closely the shape of one-factor residuals follows the

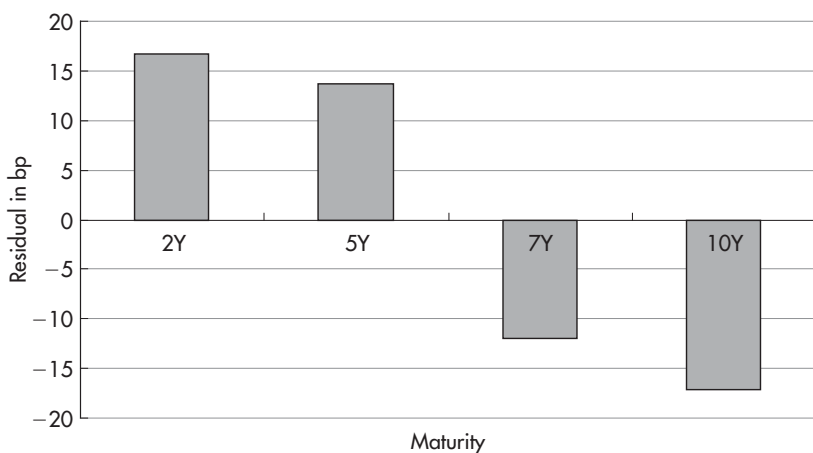


FIGURE 3.17 One-factor residuals.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data; current residuals as of 3 Oct 2011.

¹⁷Of course, he could also receive swaps instead, or buy futures, or calls on futures, etc. We shall discuss asset selection among different classes of instruments in the following chapters.

shape of the second eigenvector.) A residual of 17 bp for 2Y means that a 2Y Bund trades 17 bp too cheaply relative to the overall yield level (factor 1). These 17 bp are a source of profit uncorrelated to the direction (i.e. alpha), enhance the return in case of the bet on direction turning out to be right, and serve as a cushion in case of the bullish view turning out to be wrong.

Of course, it is also possible to trade only the residuals, without exposure to the factors. In the case of a three-factor model, this corresponds in our framework to taking no view on any of the three factors, hence hedging against all of them (by using a combination of four instruments, what is sometimes called a *condor*), leaving exposure only to the residuals. As we have seen already, the influence of economic variables is typically concentrated on the lower factors; hedging against all factors and just exploiting the residuals (i.e. the combination of all higher “factors”) usually leads to “pure” relative value trading with a focus on statistical properties.¹⁸

EXAMPLE OF A PCA-BASED TRADE IDEA

Let us now put together the elements developed above to illustrate the flow of analysis that could lead to a PCA-based trade recommendation. For the sake of simplicity, we present it in the form of a step-by-step guide (Box 3.1), but note that this cooking recipe is only a rough template that may need to be adjusted to different situations. Also, a reader wishing to get started with PCA would do well to set up an IT environment that was able to reproduce the charts used here. To facilitate this task, we have put the Excel sheet that produced these screenshots on the website accompanying the book, while referring to the disclaimer and warnings contained in it.

As the example is taken from government bond markets, the PCA is applied in combination with the fitted curve model described in Chapter 8, both as origin of the input data (constant maturity time series) into the PCA and for selecting the specific bonds in the maturity segments identified by PCA. Moreover, mean reversion models from Chapter 2 are used. While the focus of this step-by-step guide is on the contributions of PCA, Chapter 9 will embed it into a more complete analytic process for government bond markets.

¹⁸Any correlation a high factor (e.g. factor 7) might have to an external economic variable is likely to disappear in the combination of the high factors (e.g. factors 4–10) to a residual.

Box 3.1

Constructing Trade Ideas with PCA: A Step-by-Step Guide

Step 1: Decide on the relevant input data for the PCA, depending on the goals of the analysis: time horizon, type and number of variables, change or level data. Usually, at least one year of data is advisable. In our example, we use weekly level data for 2Y, 5Y, 7Y, and 10Y Bund yields from 4 Jan 2010 to 3 Oct 2011. In order to avoid rolldown effects and distortions from benchmark premia etc., constant maturity time series from a fitted curve model as described in Chapter 8 are the preferred choice for input data.

For heuristic purposes (i.e. to find the best instruments), use a large number of variables (e.g. all yields from 1 to 30 years); if the instruments of the trade are already known, restrict the input to these. Actually, one can run the PCA twice, first with a large set of variables (like from 1 to 30 years), and after the best instruments have been found for a second time (e.g. just with 2Y, 5Y, and 7Y for a butterfly trade), in particular to calculate the hedge ratios.

Step 2: Run the PCA. Check numerical stability and the results: are the calculated eigenvectors really eigenvectors and orthogonal to each other?

Step 3: Display the eigenvalues (Figure 3.4) and assess the factor structure of the market analyzed. In particular, does factorization makes sense at all (i.e. do the eigenvalues exhibit a clear factor structure)? A strong decrease of eigenvalues (i.e. $|\lambda_1| \gg |\lambda_2| \gg \dots \gg |\lambda_n|$) together with a high correlation within the data, corresponds to a clear factor structure and allows a meaningful reduction in the dimensionality of the data, of information into factors.

On the other hand, if correlation within the data is small, meaningful information reduction will be impossible. For example, if $Cov = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{pmatrix}$, then every point on the yield curve will have its own factor.

Step 4: Display the (relevant) eigenvectors (Figure 3.5) and interpret them.

Step 5: Display the time series of the (relevant) factors (Figure 3.13). This is a crucial result that serves as the basis for a number of subsequent actions, for example:

- Assess the statistical qualities of each factor; in particular, is it close to its mean or far away?
- Check the correlation between factors over subperiods. We discuss this issue in detail later.

- Use the time series of a factor as input into an Ornstein–Uhlenbeck (OU) process, for example, to assess its speed of mean reversion.
- Use that time series as explaining variable in regressions, as in Figure 3.14, in order to check the exposure of a certain position. In particular, after a trade idea has been formulated, run those regressions in order to confirm that it really has the desired exposure (especially that an RV trade is really non-directional).
- Use that time series as dependent variable in regressions versus external (candidates for) explaining variables. This is the link of the statistical analysis to the fundamental and structural analysis, which allows complementing backward-looking statistics with forward-looking expectations about macroeconomic events and potential risks. For the current example, we have provided that analysis in Table 3.1.

Step 6: Based on statistical, fundamental/structural and flow/other considerations, decide on which factor you want to take a view. In the example above and restricting ourselves to statistical reasons only, we might conclude that factor 1 has too little speed of mean reversion, thus take no view. Factor 2 is close to its mean, thus we take no view. Factor 3, however, seems to be significantly away from its mean and to have a high speed of mean reversion. Hence, we decide to investigate further for trade ideas based on factor 3 (i.e. butterflies hedged against factors 1 and 2).

Step 7: In order to select the best points on the yield curve to express that view, display the relevant residual chart, in our example, the two-factor residual chart (Figure 3.15). Decide on the maturities.¹⁹ In our example, we may want to choose sell 2Y, buy 5Y, sell 7Y (10Y has a little more negative residual than 7Y, but we might decide that the additional 0.5 bp residual is not worth the risk of going out three years further on the yield curve).

Step 8: Calculate the hedge ratios. In our example, we achieve factor 1 and 2 neutrality (i.e. the exposure to the time series of Figure 3.18) by selling 80m 2Y, buying 100m 5Y, and selling 58m 7Y Bunds. This has been calculated by a matrix inversion as discussed above. The numbers worked out as follows: with a BPV for 2Y of 1.98, for 5Y of 4.86, and for 7Y of 6.62 and sensitivities of 0.42, 0.55, and 0.53 to the first and of -0.84 , -0.04 , and 0.29 to the second factor of 2Y, 5Y, and 7Y (these numbers are the entries of the eigenvectors, which can be taken from the sheet on the website),

¹⁹ Again, other considerations than statistics play a major role, too, in particular, flow information and carry/roll-down concerns.

the matrix $\begin{pmatrix} BPV_2 \cdot e_{12} & BPV_7 \cdot e_{17} \\ BPV_2 \cdot e_{22} & BPV_7 \cdot e_{27} \end{pmatrix}$ is $\begin{pmatrix} 0.84 & 3.51 \\ -1.66 & 1.91 \end{pmatrix}$ and its inverse thus $\begin{pmatrix} 0.26 & -0.47 \\ 0.22 & 0.11 \end{pmatrix}$. Multiplying to that matrix the vector $\begin{pmatrix} -n_5 \cdot BPV_5 \cdot e_{15} \\ -n_5 \cdot BPV_5 \cdot e_{25} \end{pmatrix}$ (with n_5 to be assumed to be one), that is $\begin{pmatrix} -2.69 \\ 0.22 \end{pmatrix}$ gives the vector of weights for 2Y and 7Y: $\begin{pmatrix} -0.80 \\ -0.58 \end{pmatrix}$. Then, the relative weights can be scaled to the desired trade size, in the example above by multiplying with 100m.

Step 9: Calculate and display the time series of the butterfly using these hedge ratios (Figure 3.18). This represents the actual performance of the trade (if executed with constant maturity time series) in the past and the future performance will depend on that series. Now, run the regressions from Figure 3.14 in order to check that the individual selection of maturities offers the desired factor exposure.²⁰

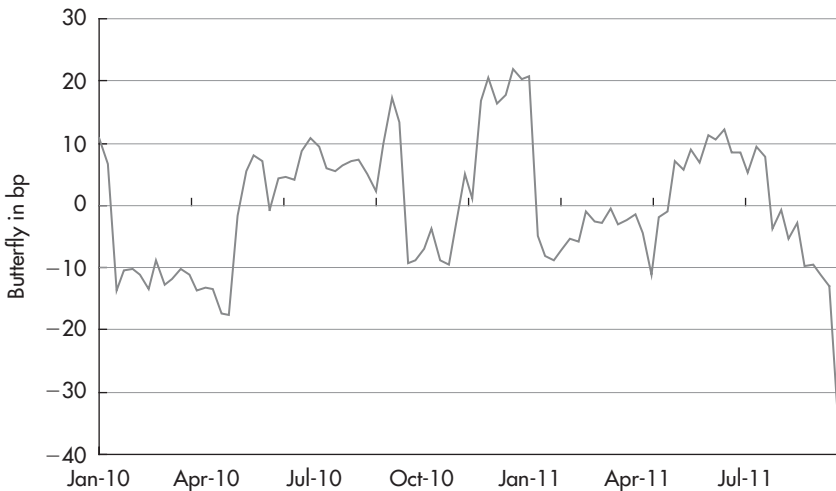


FIGURE 3.18 PCA-neutral 2Y-5Y-7Y butterfly.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

²⁰It could happen, for example, that the sensitivities to factor 3 of all the specific instruments selected are zero and that therefore the performance of the specific butterfly is uncorrelated to factor 3 and rather a function of factor 4.

Step 10: Run the OU model for the time series of Figure 3.18 in order to assess the expected performance of the trade. As outlined in Chapter 2, calculate:

- the expected profit. In our example: 32 bp;²¹
- the expected downside risk. In the OU framework, the stop loss could be set at the two-sigma level (Figure 3.19), resulting in a stop loss level that moves over time. In our example this approach would result in a stop loss level at -42 bp (loss of 7 bp) after one week.

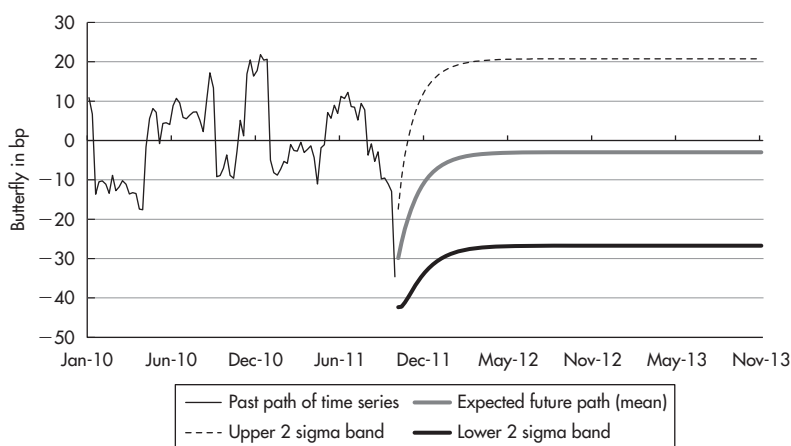


FIGURE 3.19 PCA-neutral 2Y-5Y-7Y Bund butterfly and its future path as modeled by an OU process.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data as input; forecast period as output from OU model.

- The first passage time density and the expected time until mean reversion (Figure 3.20). In our example, the trade returns to its (estimated) mean (yields 32 bp profit) on average over 84 calendar days.

²¹This chart has been generated by applying the general tool on the website, which estimates the mean. In the current case, the estimated mean of -3 bp is slightly different from the actual mean, which is 0 for any factor or residual time series from a PCA. If the actual mean is known, as in the current case, the estimation of the mean from the general tool on the website could be overwritten. Then, the expected profit would be 3 bp higher.

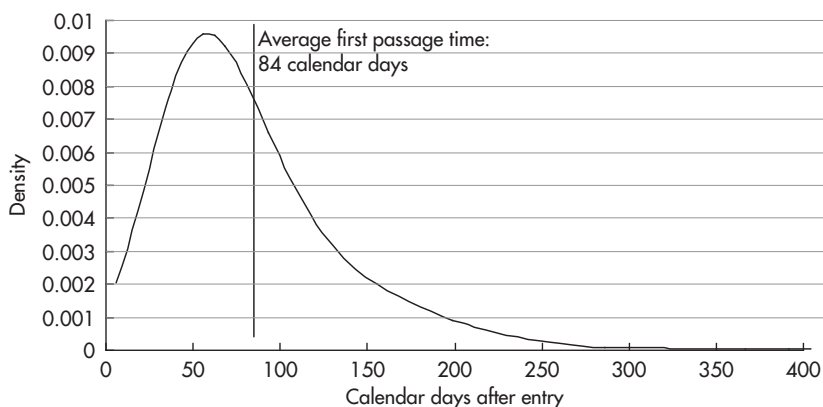


FIGURE 3.20 First passage time density for the PCA-neutral 2Y-5Y-7Y butterfly as modeled by an OU process.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data as input for the OU model.

Step 11: Select the specific bonds by using the rich/cheap information from a fitted curve model as discussed in Chapter 8. This step results in dealing from now on with two usually closely related, but still different butterflies: One with the constant maturity time series used so far, the other one with the actual bonds traded. Due to the problem of rolldown, PCA (eigenvector) and mean reversion calculations should be based on the former, as above; but the hedge ratio and hence the carry should reflect the BPV of the actual bonds traded.

Step 12: Re-calculate the hedge ratios. As the BPV for the specific bonds can be different from the BPV of the constant maturity time series, the hedge ratio calculation described above should be repeated with the BPV of the specific bonds.²² Based on the expected holding horizon

²²Ideally, one would also want to re-run the PCA with the specific bonds selected, but faces the issue of rolldown. Hence, the best compromise practically possible seems to be using the BPV of the specific bonds and the eigenvectors of the PCA on constant maturity time series.

(84 calendar days) of the OU process and the hedge ratios, calculate the carry. Note that in the absence of BPV neutrality carry cannot naturally be expressed in bp terms anymore. Therefore, calculate the carry in money terms (i.e. euros and cents), based on the PCA-neutral hedge ratios. In a PCA framework, it is natural to express carry in money terms; should expression in bp terms be required, it must be stated, to *which* maturity the bps refer (as PCA does away with the false assumption of parallel curve shifts). In our example, the 84 calendar day carry is EUR 70,000 positive,²³ thus rather negligible compared to the profit potential from mean reversion. If the carry had been significantly negative, calculating additionally the carry until the 90% quartile of Figure 3.20 could have helped assessing whether the trade was still attractive in case it took unusually long to revert back to its mean.²⁴

And how did the trade in Box 3.1 perform in reality? Figure 3.21 compares the actual evolution with the forecast of the OU model. It may be no surprise to the reader that we have picked an example that worked well. In the following we will discuss instances where PCA might not work that smoothly, categorizing the problems of PCA into:

- Correlation between factors during subperiods. This mainly affects short-term trades, like the RV position just described.
- Instability of the eigenvectors. This mainly affects long-term trades, such as curve positions motivated by cyclical macro-economic considerations.

²³As this is for illustration purposes only, we have used general collateral (GC) rates for the carry calculation.

²⁴Imagine a trade with a 10 bp profit potential and a three-month negative carry also of 10 bp. If the OU model shows an expected holding horizon of just two weeks (over which the negative carry is, say, 2 bp), the negative carry might be acceptable. The confidence in this position could increase further if the OU process suggests a 90% chance of the 10 bp profit to materialize over one month. Alternatively, one could calculate the probability of the mean reversion taking longer than three months (i.e. the likelihood that the negative carry would exceed the profit from mean reversion).

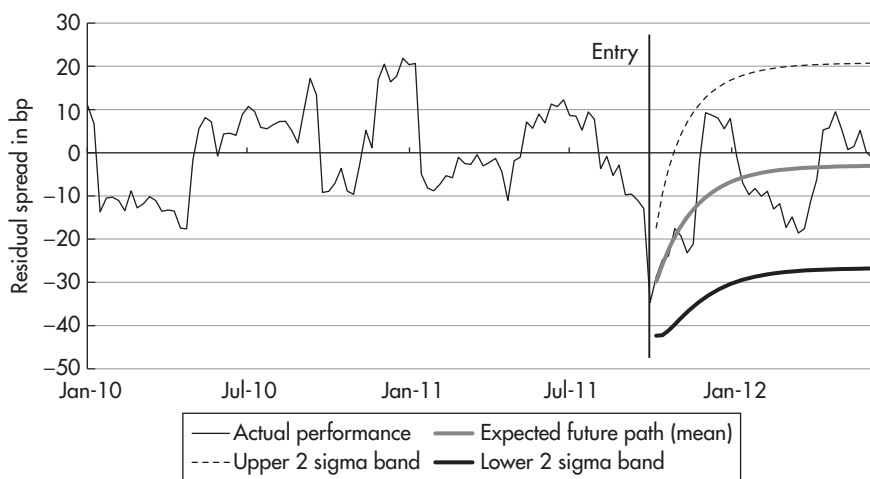


FIGURE 3.21 Performance of Bund butterfly after entry compared to OU model forecast.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 11 Jun 2012, weekly data.

PROBLEMS AND PITFALLS OF PCA 1: CORRELATION BETWEEN FACTORS DURING SUBPERIODS

By construction, factors are uncorrelated over the whole sample period used as input data into a PCA, which allows one to construct a position hedged against certain factors. However, there could occur correlation between factors during subperiods. In such a case, the hedge could break down during that subperiod. For example, a trade on factor 2 hedged against factor 1 would be exposed to factor 1 during a time period, in which ephemeral correlation between factor 1 and factor 2 occurred. In this case, the performance of what was intended to be a non-directional steepening position would be driven (ephemerally) by direction.

To see how the theoretical problem of correlation between factors during subperiods can affect trades, let's consider the example of a 2Y-10Y PCA-neutral steepening trade on the Bund curve. In October 2010 and October 2011, the time series shown in Figure 3.22 may well have looked too good to resist from entering the trade.

However, before hitting the target (reverting back to mean), a steepening position entered in October 2011 broke through the OU stop loss and subsequently underperformed further. The reason for this misbehavior can be seen in the graph in Figure 3.23, which regresses the residual spread (PCA-neutral steepening trade) against the first factor, just as in Figure 3.14.



FIGURE 3.22 PCA-neutral 2Y-10Y Bund steepening position.

Sources: data – Bloomberg; chart – Authors.

Data period: 7 Jan 2008 to 11 Jun 2012, weekly data.

It turns out that, while overall the correlation is zero, there has been a high correlation to the first factor on the left-hand side of the chart in Figure 3.23. Further investigation reveals that in the six months before entry the trade (a factor 2 position) followed factor 1 very closely. Thus, the reason for the underperformance of the PCA-neutral Bund steepening trade was that it was in fact a directional trade, and, since the direction happened to work against it, its less-strong factor 2 exposure could not save it from losing.²⁵ Compare this situation with the regressions depicted in Figure 3.14, where we have argued against BPV-neutral weighting due to its factor 1 exposure: in times of ephemeral correlation between the factors, a PCA-neutral trade shares the fate and criticism of BPV-neutral trades.

The good news is that in many cases correlation between factors during subperiods can be spotted before entering into a trade. To do so, it is crucial to run the regressions (step 5 of the flow above). Often, a potential problem with factor correlation announces itself, as in the current example. Thus, if there has been a high correlation between the factors in the time period before (as in Figure 3.23), caution is advisable. By contrast, the 2Y-5Y-7Y butterfly

²⁵Also in October 2010, there was a high correlation to factor 1, but this time the direction worked in favor of the steepening trade. Thus, in 2010, the trade made money, but for the wrong reason: not because it was a factor 2 position hedged against factor 1 as intended but because we were lucky that the direction worked in our favor.